Weight Initialisation: What Not to Do

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Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
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- **Builds on:** Backpropagation; Exploding gradient and gradient clipping; Vanishing gradient; Improving a neural network.
- **Leads to:** Xavier and He initialisation.

1. Overview

Key point: The starting weights decide whether a network trains at all: all zeros, one shared constant, very small and very large random numbers all fail. The weights must be random, with a spread that is neither too small nor too large.

Training a network begins with one step before the loop: giving every weight and bias a starting value, its **initialisation** (G-947). A bad start can cause three problems:

- the **vanishing gradient** (G-2070) problem: the early layers stop learning;
- the **exploding gradient** (G-731) problem: updates become huge and erratic;
- **slow convergence** (G-472): the network gets to a good solution only after very many **epochs** (G-696).

This Note tries the four bad starts one by one, each in Keras or numpy, and shows which problem it causes. The good starts, Xavier and He, are in Xavier and He initialisation.

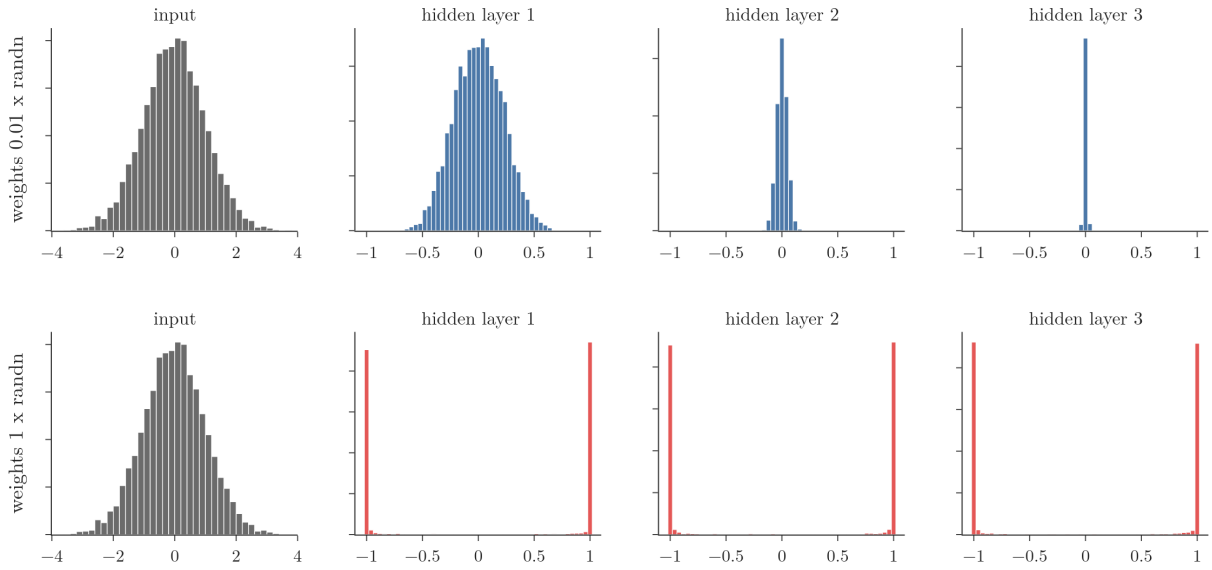


Figure 1: Activations of a tanh network with three hidden layers of 500 nodes. Top: weights 0.01 × standard normal; the values shrink towards 0 layer after layer. Bottom: weights 1 × standard normal; almost every value is stuck at −1 or 1

Figure 1 shows the two random starts that fail: too small, and the signal dies out; too large, and every node saturates.

2. Prerequisites

- The steps of backpropagation: the training loop, and the Extra on nodes that start equal (one update).
- The vanishing gradient problem and the exploding gradient problem.
- Activation functions: sigmoid, tanh, **ReLU** (G-1668) and saturation (a function's flat ends).

3. Why the starting weights matter

> **Key point:** Initialisation is step 0 of training, and it alone can decide between learning and not learning. Bad starts, with the **sigmoid** (G-1798), stalled deep learning research for years.

Training a network repeats four steps after the start (see the steps of backpropagation):

0. **Initialise** every weight and bias, and choose an **optimizer** (G-1401).
1. **Forward propagation** (G-797): compute $\hat{y}$ for an input.
2. **Loss:** compare $\hat{y}$ with y.
3. **Gradients:** compute the derivative of the loss for every parameter.
4. **Update:** move every parameter against its gradient.

Steps 1 to 4 all start from what step 0 chose. When researchers in the 1990s and 2000s finally understood why deep networks would not train, they found two culprits behind the **vanishing gradient** (G-2070): the sigmoid activation and badly chosen starting weights (Glorot and Bengio 2010). Research on initialisation has continued ever since.

4. Do not start every weight at zero

> **Key point:** With **ReLU** (G-1668) or tanh, all-zero weights never change: training does nothing. With sigmoid the weights change, but every node in a layer stays identical, so the layer acts like one node.

4.1 The setup

> **Key point:** Two inputs (CGPA, IQ), one hidden layer of two nodes, one output node; every weight and bias starts at 0.

Take a regression problem:

- each **observation** (G-1374; one record) is a student;
- the two **features** (G-772; input variables) are CGPA x_1 and IQ x_2;
- the **target** (G-1949; the output we predict) is the placement package in lakhs.

The network has two hidden nodes and one linear output node. Write W^1_{ij} for the weight from input i to hidden node j, so the hidden nodes compute

$$z_{11} = W^1_{11}x_1 + W^1_{21}x_2 + b_{11}$$

$$z_{12} = W^1_{12}x_1 + W^1_{22}x_2 + b_{12}$$

and pass each sum through the activation function g:

$$a_{11} = g(z_{11})$$

$$a_{12} = g(z_{12})$$

With every weight and bias at 0, both z are 0 for every student.

4.2 ReLU and tanh: nothing moves

> **Key point:** Both give $a = 0$ at $z = 0$. Every gradient into or out of the hidden layer then contains a 0 factor, so no weight is ever updated.

With **ReLU** (G-1668), $a_{11} = \max(0, 0) = 0$; with **tanh** (G-1947), $a_{11} = \tanh(0)$, and

$$\tanh(0) = \frac{e^0 - e^{-0}}{e^0 + e^{-0}} = \frac{1-1}{1+1} = 0$$

The same holds for a_{12}. Now look at the gradients:

1. **In words:** a weight's gradient contains the signal flowing in (an activation or input) and the signal flowing back (through the weights after it). Here one of the two is always 0.
2. **Formula:** for an output weight and a hidden weight,

$$\frac{\partial L}{\partial W^2_{11}} = \frac{\partial L}{\partial \hat{y}}\, a_{11}$$

$$\frac{\partial L}{\partial W^1_{11}} = \frac{\partial L}{\partial \hat{y}}\, W^2_{11}\, g'(z_{11})\, x_1$$

   Here $g'(z_{11})$ is the slope of the activation function g at z_{11} (for a **sigmoid** (G-1798), at most 0.25).
3. **Example:** $a_{11} = 0$ makes the first gradient 0, and $W^2_{11} = 0$ makes the second 0, whatever $\partial L/\partial \hat{y}$ and x_1 are. So with **learning rate** (G-1068) η the update leaves the weight at 0:

$$W_{\text{new}} = W_{\text{old}} - \eta \cdot 0 = 0$$

The weights are 0 after the first update, so the same happens again, forever. Only the output bias, whose gradient is just $\partial L/\partial \hat{y}$, can move. The network predicts one constant for every input.

We check this in Keras on 300 observations of `make_moons` (G-111; two features, two classes), with two ReLU hidden nodes and a sigmoid output. We overwrite Keras' starting weights with zeros and train for 100 **epochs** (G-696):

- every hidden and output weight is still exactly 0;
- only the output bias moved, to -0.0024;
- every observation gets the same prediction, so the network is no better than guessing on these 150 + 150 points (Keras reports 47%, an average over the last epoch while the bias still wobbles around 0).

Tanh gives the same numbers.

> **Python:** Overwriting the starting weights.
> ```
> weights = model.get_weights()    # list of arrays
> zeros = [np.zeros_like(w) for w in weights]
> model.set_weights(zeros)
> ```
> `get_weights()` returns every weight matrix and bias vector in order; `set_weights()` replaces them with arrays of the same shapes.

4.3 Sigmoid: every node becomes the same node

> **Key point:** $\sigma(0) = 0.5$, so the weights do move, but nodes that start equal get equal gradients and stay equal. Ten hidden nodes act like one, and the decision boundary is a straight line.

With the **sigmoid** (G-1798), $a_{11} = a_{12} = \sigma(0) = 0.5$. The activations are no longer 0, so the output weights get non-zero gradients and training starts. The problem is that the two activations are equal, and stay equal.

Compare the gradients of the two weights leaving x_1, by the **chain rule** (G-371):

$$\frac{\partial L}{\partial W^1_{11}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_{11}} \cdot \frac{\partial a_{11}}{\partial z_{11}} \cdot x_1$$

$$\frac{\partial L}{\partial W^1_{12}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_{12}} \cdot \frac{\partial a_{12}}{\partial z_{12}} \cdot x_1$$

- The first factor is shared.
- The second factors, $\partial \hat{y}/\partial a_{11}$ and $\partial \hat{y}/\partial a_{12}$, are the output weights W^2_{11} and W^2_{12}, which start equal and receive equal updates.
- $z_{11} = z_{12}$, so the sigmoid slopes are equal.
- The last factor is x_1 in both.

So the two gradients are equal at every step. Think of two identical twins who sit the same lessons, hear the same feedback and make the same corrections: nothing ever makes them differ. The two weights from x_1 start equal and move together; so do the two weights from x_2. Equal gradients for equal weights are the symmetry (the **symmetry problem**, G-1933) noted in one update with real numbers: nodes that start identical stay identical.

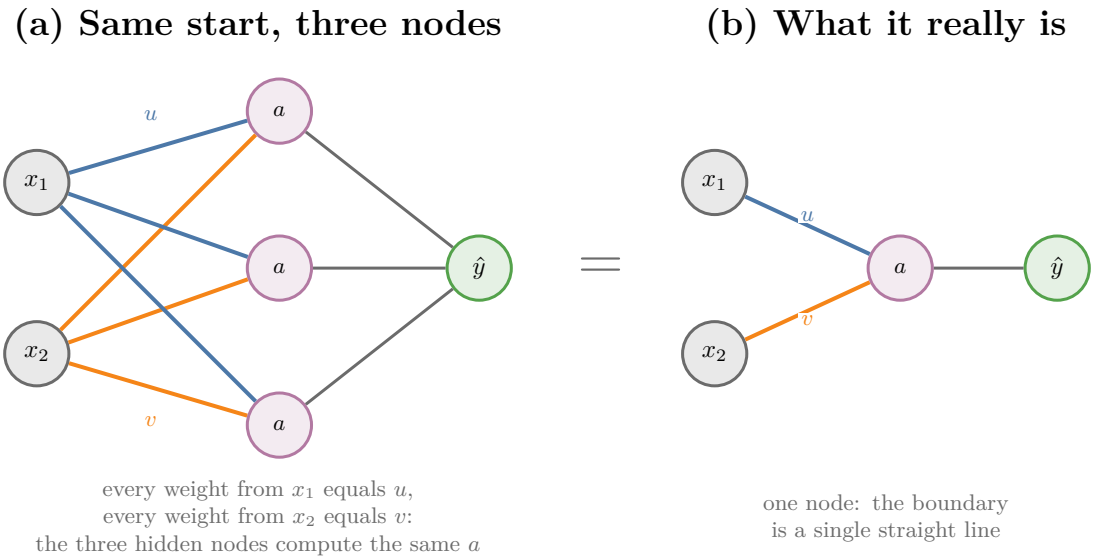


Figure 2: (a) After a start where all weights are equal, every weight leaving x_1 has one value u and every weight leaving x_2 one value v, so the hidden nodes all compute the same a. (b) The network behaves like a single hidden node

Figure 2 shows the consequence. However many nodes the layer has, they compute the same thing, so the network behaves like one with a single hidden node. A single sigmoid node has a straight **decision boundary** (G-555), the line where the predicted class changes:

1. the hidden node and the output compute

$$a = \sigma(ux_1 + vx_2 + c)$$

$$\hat{y} = \sigma(w\,a + b)$$

   and $\hat{y}$ only grows or only shrinks as $ux_1 + vx_2$ grows;
2. so the output crosses 0.5 where $ux_1 + vx_2$ equals one constant;
3. that set of points, $ux_1 + vx_2 =$ constant, is a straight line, and it is the decision boundary.

The network is in effect a **perceptron** (G-1486): a linear model that cannot capture non-linear patterns.

The Notebook trains a layer of 10 sigmoid nodes from all zeros for 200 **epochs** (G-696). Afterwards:

- all 10 weights from x_1 are 0.436, and all 10 weights from x_2 are -1.186;
- all 10 biases are -0.822;
- the decision boundary is a straight line, with 87% accuracy (Figure 3, left).

The same network with Keras' random start bends around the moons and reaches 96% (Figure 3, right).

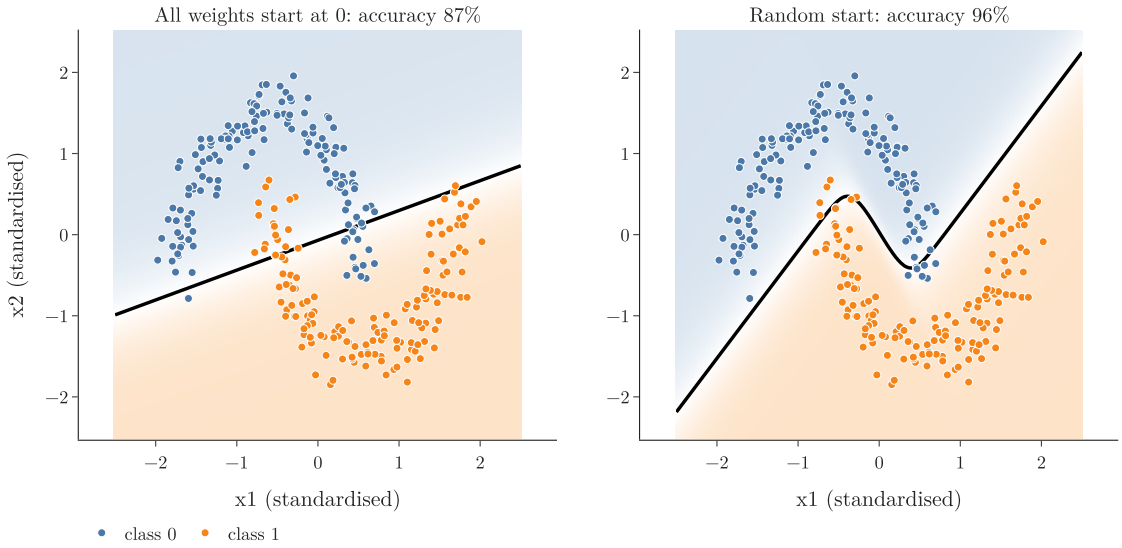


Figure 3: The same 10-node sigmoid network after 200 epochs. Left: started from zeros, it draws a straight line. Right: started from random weights, it bends around the moons

In Figure 3, the background shade is the probability the network gives to class 1 at that spot: light orange where it favours class 1, light blue where it favours class 0, white at 0.5. The black line is the decision boundary, where the probability is 0.5. The dots are the training points (blue: class 0, orange: class 1).

5. Do not start every weight at the same non-zero value

> **Key point:** Starting all weights at 0.5 avoids the zeros but not the symmetry: with **ReLU** (G-1668), tanh or sigmoid, every node in a layer stays identical and the network stays linear.

Starting every weight and bias at a non-zero constant, say 0.5, removes the zeros of section 4.2. With ReLU,

$$z_{11} = 0.5\,x_1 + 0.5\,x_2 + 0.5 = z_{12}$$

is now some non-zero value, so $a_{11} = a_{12} \neq 0$. But the two are equal, which is exactly the situation of section 4.3. Every weight leaving one input gets the same gradient, the nodes

stay identical, and the layer acts like one node.

The stuck classifier in training the classifier, where every weight starts at 0.1, is this same problem.

The Notebook confirms it with 3 hidden nodes and 100 **epochs** (G-696):

Activation	Weights from x_1	Weights from x_2	Accuracy
ReLU	0.312, 0.312, 0.312	−0.842, −0.842, −0.842	88%
Tanh	0.192, 0.192, 0.192	−0.544, −0.544, −0.544	88%
Sigmoid	0.744, 0.744, 0.744	−2.062, −2.062, −2.062	88%

Figure 4 follows the ReLU row of the table through training, next to the same network with Keras' random start.

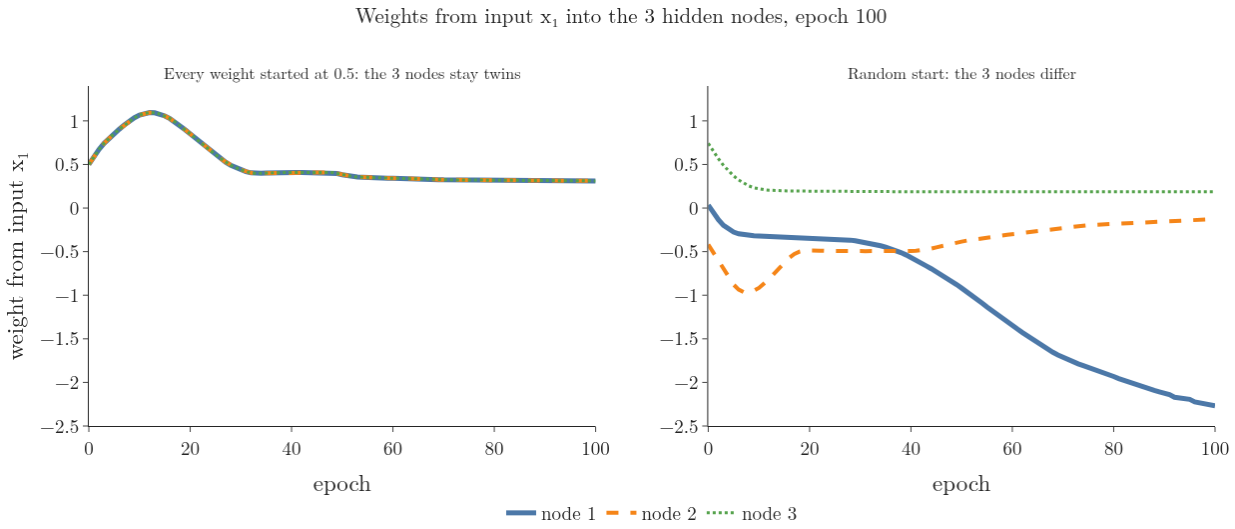


Figure 4: The weights from input x1 into the 3 ReLU hidden nodes during 100 epochs. Left: every weight and bias started at 0.5. Right: Keras' random start.

- **Started at 0.5 (left).** The three weights rise to about 1, fall back and settle at 0.312. They move as one line from the first epoch to the last, because at every step the three nodes receive the same gradient.
- **Random start (right).** The three weights start at different values, receive different gradients and end at −2.27, −0.13 and 0.19: three nodes doing three different jobs.

The network trains, but only as a linear model. Sections 4 and 5 together lead to one rule: every weight must start at a different value. The way to get that is to draw the starting weights at random.

6. Do not use very small random weights

> **Key point:** Random weights of about ±0.01 make every weighted sum tiny. With tanh and sigmoid the gradients vanish; with **ReLU** (G-1668) the network converges extremely slowly.

6.1 The signal shrinks layer by layer

> **Key point:** 500 inputs times weights of about 0.01 give z close to 0. **Tanh** (G-1947) keeps it close to 0, and each layer shrinks it further: the standard deviation goes 0.21, 0.048, 0.011.

Take a dataset with 1000 observations and 500 features, each feature a standardised number (mean 0, **standard deviation** (G-1871) 1). The network has three hidden layers of 500 nodes. Every weight is drawn as `np.random.randn(500, 500) * 0.01`, a **standard normal** (G-1873) number times 0.01, and every bias is 0.

Each node computes $z = \sum_{i=1}^{500} w_i x_i$. With inputs around ±1 and weights around ±0.01, the products are tiny and partly cancel, so z is small. Tanh of a small number is about the number itself, so the activations are close to 0.

Figure 1 (top row) shows the **histograms** (G-899): each bar's height is how many of the activations fall in that small range of values, so a narrow spike means nearly all values are the same. The input has standard deviation 1; the activations of hidden layers 1, 2 and 3 have standard deviations 0.21, 0.048 and 0.011. In layer 3, 66% of the activations lie between −0.01 and 0.01.

The next layer multiplies these small activations by small weights again, so every layer is smaller than the one before. During backpropagation many such small numbers multiply into the gradients, and the gradients of the early layers become close to 0. The result is the vanishing gradient problem, caused here by the start alone.

6.2 The three activations

> **Key point:** Tanh: strong vanishing gradients. Sigmoid: weaker, but training is very slow. **ReLU** (G-1668): no saturation, but convergence is extremely slow.

- **Tanh:** the activations collapse towards 0, as just shown; the gradients vanish strongly, and the weights may not update at all.
- **Sigmoid:** $\sigma(0) = 0.5$, so the activations cluster around 0.5 (**standard deviation** (G-1871) 0.055 in layer 1, then 0.027 in layers 2 and 3), not around 0. The vanishing gradient is weaker than with tanh, but training is very slow or stalls.
- **ReLU:** the activations also shrink (standard deviation 0.131, 0.021, 0.003), though ReLU does not squash them. The gradients do not vanish as strongly, but **convergence** (G-472) is extremely slow: in section 6.3 a small ReLU network started this way does not move in 100 epochs.

The problem gets worse with depth. A shallow network has few factors to multiply; a deep one has many.

6.3 In Keras: the weights do not move

> **Key point:** Four hidden layers of 10 nodes, weights 0.01 × **standard normal** (G-1873), plain SGD: the loss stays at 0.6934 for 100 epochs and the weights match their start to four decimals. Keras' default start reaches a loss of 0.002.

The Notebook builds four hidden layers of 10 nodes on the moons data, with tanh or **ReLU** (G-1668), starts every weight at 0.01 times a standard normal number and every bias at 0, and trains with plain **SGD** (G-1892; **learning rate** (G-1068) 0.1) for 100 **epochs** (G-696):

- **Tanh:** the loss is 0.6934 at the start and 0.6934 at the end, the loss of guessing ($\ln 2 = 0.693$). The first-layer weights read $0.0035, 0.0082, 0.0033, -0.0130$ before and after, identical to four decimals. Accuracy 47%, no better than guessing.
- **ReLU:** exactly the same picture.
- **Keras' default start,** same networks: the loss falls to 0.002 (**tanh** (G-1947)) and 0.001 (ReLU), with 100% accuracy.

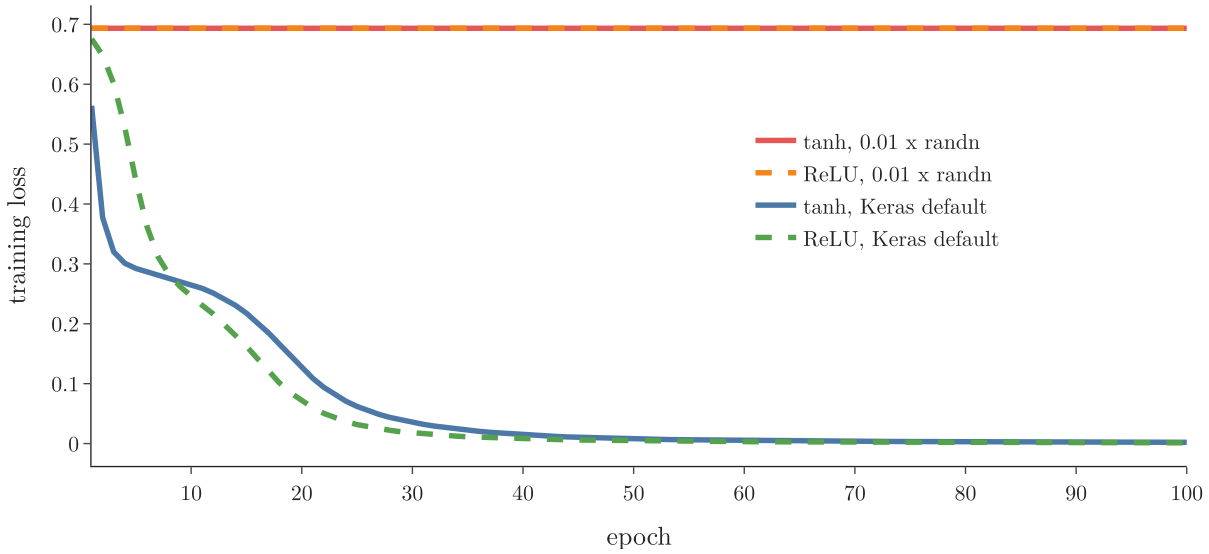


Figure 5: Training loss of the four-layer network with plain SGD. Started from 0.01 × standard normal, tanh and ReLU stay at 0.693 for 100 epochs. Started from Keras' default, both fall to near 0

Figure 5 shows the four loss curves. The only difference between the flat and the falling ones is the starting weights.

> **Extra:** With the **Adam** (G-169) **optimizer** (G-1401) (learning rate 0.01) instead of plain SGD, the tanh network does learn (loss 0.27, accuracy 87%) while the ReLU network

stays stuck at 0.693. Adam divides each step by the recent size of that weight's gradients, so the step size does not depend on how small the gradients are (Kingma and Ba 2015). Adam therefore hides the vanishing gradient in the weights, as the Extra on gradients and weights explains, but does not always rescue training.

7. Do not use large random weights

Key point: Weights around ±1 make z a sum of 500 sizeable terms. Tanh and sigmoid saturate, causing slow training or vanishing gradients; **ReLU** (G-1668) passes the large values on, causing exploding gradients and unstable training.

Now draw the weights without the factor 0.01: `np.random.randn(500, 500)`, mostly between −3 and 3. (Uniform numbers between 0 and 1 cause the same problem: in the Notebook, 84%, 99% and 100% of the **tanh** (G-1947) outputs of layers 1, 2 and 3 lie beyond ±0.99.) In deep learning these count as large weights.

1. **In words:** z adds up 500 products of inputs around ±1 and weights around ±1.
2. **Formula:** $z = \sum_{i=1}^{500} w_i x_i$
3. **Example:** if each product were about 0.5 and they did not cancel, z would be about

$$500 \times 0.5 = 250$$

With cancellation, the **variance** (G-2074; the average squared distance from the mean, written Var) of z follows this rule (derived in the variance of z):

$$\mathrm{Var}(z) = n \,\mathrm{Var}(w)\,\mathrm{Var}(x)$$

Here $n = 500$ inputs, and `randn` weights and the inputs both have variance 1:

$$\mathrm{Var}(z) = 500 \times 1 \times 1 = 500$$

$$\sqrt{500} \approx 22 \quad \text{(the standard deviation)}$$

So z typically lands in the tens.

Tanh and sigmoid saturate. Feeding z of 20 or 50 into tanh gives −1 or 1; into the **sigmoid** (G-1798), 0 or 1. Figure 1 (bottom row) shows it: in every hidden layer the tanh activations pile up at −1 and 1. In layer 3, 90% of them are beyond ±0.99. At these values the slope is almost 0, so training is slow at best, and in the worst case the gradient vanishes.

ReLU explodes. ReLU does not saturate on the positive side: if $z = 250$, the output is 250. The large values pass on and grow. In the Notebook the mean ReLU activation is 8.9 in layer 1, 140 in layer 2 and 2,224 in layer 3. The chain that follows:

1. large activations give large gradients;
2. large gradients give huge jumps in **gradient descent** (G-862);
3. the weights swing back and forth without settling.

This is the exploding gradient problem.

Figure 6 follows the tanh signal through 10 layers of the same wide network instead of 3. Watch the three rows split apart:

- with 0.01× randn, the values close in on 0, about 4 times tighter at every layer;
- with 1× randn, they sit at −1 and 1 from the first layer on;
- only the middle spread, randn/$\sqrt{500}$, keeps them in between.

That middle start is **Xavier initialisation** (G-2131), derived in Xavier initialisation.

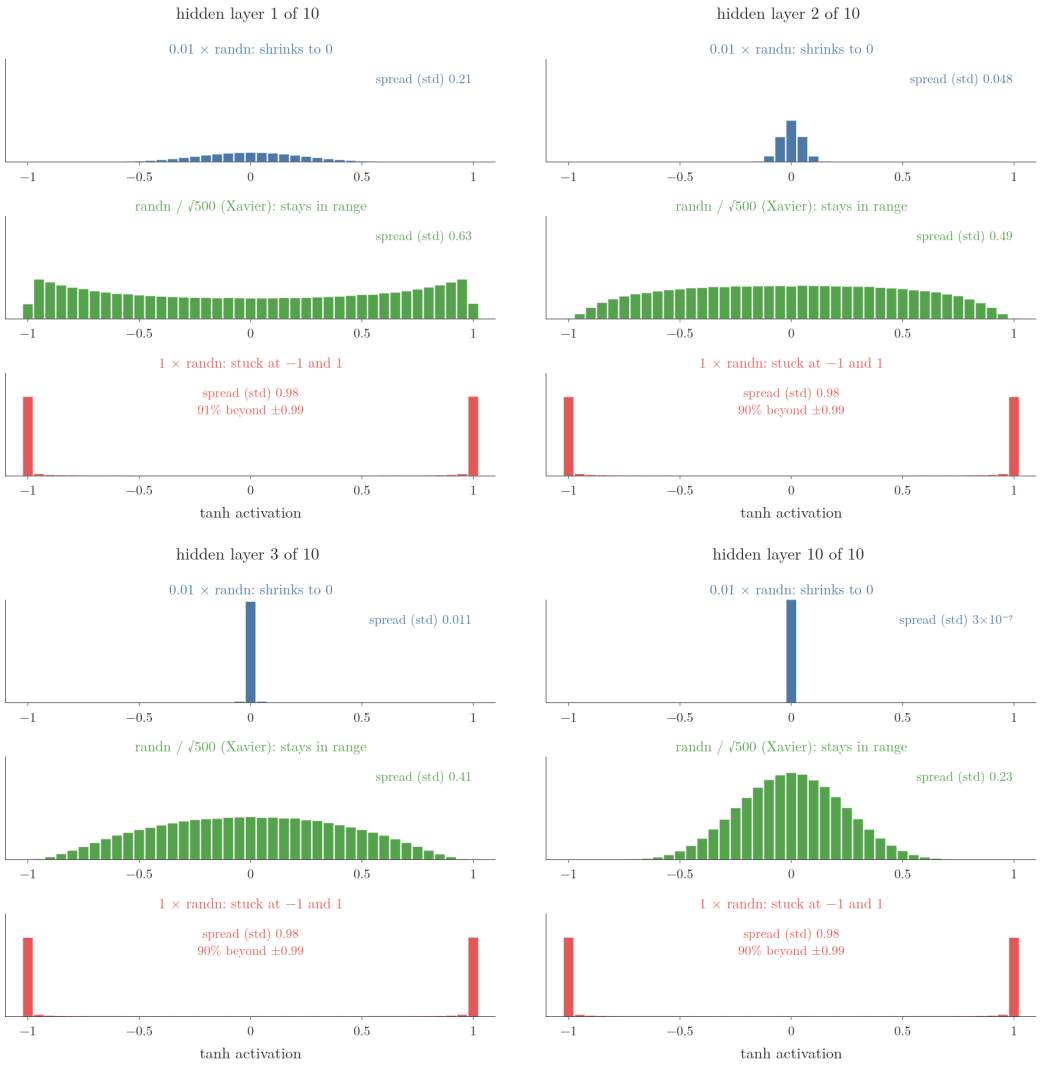


Figure 6: Tanh activations of a network with 10 hidden layers of 500 nodes, one layer per frame, for three starting spreads. Blue: 0.01× randn, the spread falls from 0.21 to 3×10^{-7}. Green: randn/$\sqrt{500}$, it stays between 0.63 and 0.23. Red: 1× randn, about 90 percent of the values lie beyond ±0.99 in every layer. Each row has its own height scale

8. Summary

Start	ReLU	Tanh	Sigmoid
All zeros	no training (weights stay 0)	no training (weights stay 0)	nodes stay identical: linear model
One constant (0.5)	nodes identical: linear model	nodes identical: linear model	nodes identical: linear model
Small random (0.01× randn)	extremely slow convergence	vanishing gradient	vanishing gradient, very slow
Large random (1× randn)	exploding gradient, unstable	saturation: slow or vanishing	saturation: slow or vanishing

- Weights must start different from each other, because nodes that start the same get the same gradient and stay identical, so they must be random.
- Their spread must be neither too small (the signal dies out) nor too large (it saturates or explodes).
- In the moons experiments: zeros leave every weight at 0 (47%); a **sigmoid** (G-1798) layer started at zeros acts as one node (87%, straight line); 0.01-scale weights keep the loss at 0.693 with plain SGD.
- Choosing that spread well is the job of Xavier and He **initialisation** (G-947).

9. Sources

Built from

- CampusX, "Weight Initialization Techniques | What not to do? | Deep Learning", YouTube, https://www.youtube.com/watch?v=2MSY0HwH5Ss

Other references

- Glorot, X. and Bengio, Y. (2010). Understanding the difficulty of training deep feedforward neural networks. AISTATS 2010, PMLR 9:249-256.
- Kingma, D. P. and Ba, J. (2015). Adam: A Method for Stochastic Optimization. ICLR 2015. arXiv:1412.6980.

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Initialisation (G-947)	Giving every weight and bias its starting value before training begins; a bad start can cause vanishing or exploding gradients or slow convergence.

Term	Meaning
Symmetry problem (G-1933)	Nodes that start with equal weights get equal updates and stay identical, so a layer acts like one node.
`get_weights()` / `set_weights()` (G-85)	Keras methods that read and replace the weight and bias arrays of a model or a layer, as a list of arrays.
Slow convergence (G-1824)	Reaching a good solution only after very many epochs.
Zero initialisation (G-2144)	Starting every weight (and bias) at 0; with ReLU or tanh nothing trains.
Vanishing gradient (G-2070)	Gradients shrinking towards 0 as they pass through many layers, which slows learning.
Exploding gradient (G-731)	Gradients growing huge as they pass back through many layers, making updates erratic.
Convergence (MICE) (G-472)	The point where the fills hardly change between two iterations.
Epoch (G-696)	One full update of the parameters using the whole training set.
ReLU (G-1668)	Rectified linear unit, the activation $\max(0, z)$: it passes a positive input unchanged and turns a negative one into 0; the usual default activation for hidden layers.
Sigmoid function (G-1798)	An S-shaped function that squashes any number into the range 0 to 1, $\sigma(z) = 1/(1 + e^{-z})$; it turns a score into a probability, as in logistic regression.
Optimizer (G-1401)	The rule that turns gradients into weight updates, such as plain gradient descent or Adam.
Forward propagation (G-797)	Passing one row of inputs through the network, layer by layer, to get the prediction.
Observation (G-1374)	One record of a dataset: one row of the data table.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Tanh (hyperbolic tangent) (G-1947)	An activation function that squashes any number into an S-curve from -1 to 1, centred on 0: $\tanh(z) = (e^{z} - e^{-z})/(e^{z} + e^{-z})$, with derivative $1 - \tanh^2(z)$.

Term	Meaning
Learning rate (η) (G-1068)	How strongly each update changes the model; in gradient descent, the number the slope is multiplied by to get the step size.
`make_moons` (G-111)	scikit-learn function that generates two interleaving half-moon classes.
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.
Decision boundary (G-555)	A line or curve that separates the classes in classification.
Perceptron (G-1486)	The smallest building block of a neural network, one artificial neuron: it multiplies each input by a weight, adds the results and turns the sum into an output.
Standard deviation (G-1871)	How far values typically lie from the mean; divide by $n-1$ for a sample (pandas default) or by n for a population (NumPy default).
Standard normal distribution (G-1873)	The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0,1)$; any normal variable can be converted to it with z-scores, so one table of probabilities serves every normal distribution.
Histogram (G-899)	A bar chart of how many values fall in each equal range (bin) of a numerical column.
Stochastic gradient descent (SGD) (G-1892)	Gradient descent that uses one random row for every update, so each update is cheap and many updates happen per epoch, at the cost of a noisier path.
Adam (G-169)	An optimizer, the rule that updates a network's weights to reduce the loss: a variant of gradient descent that keeps running averages of past gradients and of their squares, giving each weight its own step size; fairly robust to its settings, so a common default.
Variance (of data) (G-2074)	The average squared distance of the values from their mean; the square of the standard deviation; divide by n for a population (NumPy default) or $n-1$ for a sample (pandas default).
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.

Term	Meaning
Xavier (Glorot) normal (G-2131)	Starting weights drawn from a normal distribution whose spread is set by the layer's size, standard deviation $\sqrt{1/\text{fan-in}}$ or $\sqrt{2/(\text{fan-in} + \text{fan-out})}$, so the signal keeps its size from layer to layer.